Biochemistry Cume Sept 6, 2013

Protein Function and Stability

1. Why do proteins typically denature under acidic or basic conditions? Describe and illustrate your answer in detail. (20 pts.)

2. Draw the catalytic triad in a "Generic" serine protease. Order the residues from left to right. Show where a generic substrate is likely to be placed, next to the catalytic atom in your drawing. Show appropriate bonding for a covalent enzyme-substrate intermediate that is formed during the enzyme-catalyzed reaction. (40 pts.)

3. Define and describe in detail allosterism. (20 pts.)

4. Define and describe a zymogen. (20 pts.)